D6: The two misspecifications that actually bite

Are lag and saturation present in the real curves, and does a corrected model change anything?

OxygenModel D-series

2026-08-07

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Verdict

Neither vulnerability is present in the real curves, and the current model is adequate. No change to the default estimator follows from these numbers.
D5 measured, in simulation, that an equilibration lag biases per-cell R by +18.6% and a saturating growth phase by −78.1%. D6 asked whether either is actually there. Neither is:

- A nested F-test against each saturating alternative rejects at chance level — 3/75 curves for growth saturation, 3/75 for O limitation — and the extra term buys a median RSS reduction of 0.011%.

- **The reason is the trimming.** The median window still holds 44.5% of its starting oxygen when the fit ends, and only 7 of 75 end below 20%. O limitation cannot bite there. D5's −78.1% arm was pessimistic *about this data* because it let growth run to carrying capacity inside the window.

The window-sensitivity hypothesis fails, and the real explanation is arithmetic. All six saturation measures together explain $R^2 = 0.168$ of the per-curve movement in R. Fitted r alone explains $R^2 = 0.977$ of the movement in K — because shifting a window edge by Δ on a curve growing at rate r changes the oxygen consumed by $\approx e^{r\Delta}$, so K must move by $\approx r\Delta$. The prediction is 13.4% against an observed 15.4%.

This is "no effect", not "no power." On simulated curves carrying a planted 15-minute lag, the lag model wins on AIC in 100% of cases and recovers the lag at a median of 15 minutes — exact. On clean curves it is correctly rejected at 0.9%.

And the key test agrees. Window sensitivity does not fall under the extension: median $\max|\Delta K|$ is 14.4% under M0 against 14.5% under M2g. Between-taxon ordering of per-cell R is identical (Spearman 1.000) under every model. **No published conclusion moves.**

1 What this report does

D5 established that uncertainty propagation is not this pipeline's problem, and pointed instead at bias from misspecification. It measured, in simulation:

arm	bias in per-cell R
none	+0.09%
optode drift	−1.1%
initial lag	**+18.6%**
saturating growth	**−78.1%**

D6 asks whether lag or saturation is present in the 75 real curves, and if so what a corrected model does. It changes no pipeline number, constant or analysis decision: every model here is an **additional** fit writing to `runs/D6_analysis/`.

Every number below is rendered from a CSV in that directory. Nothing is recomputed in this document.

2 Is saturation present?

Each curve was refit over its committed window and the residuals examined against time-in-window and against absolute oxygen. The two mechanisms make different predictions — O limitation scales with absolute O , growth saturation with elapsed growth — which is what makes them separable in principle.

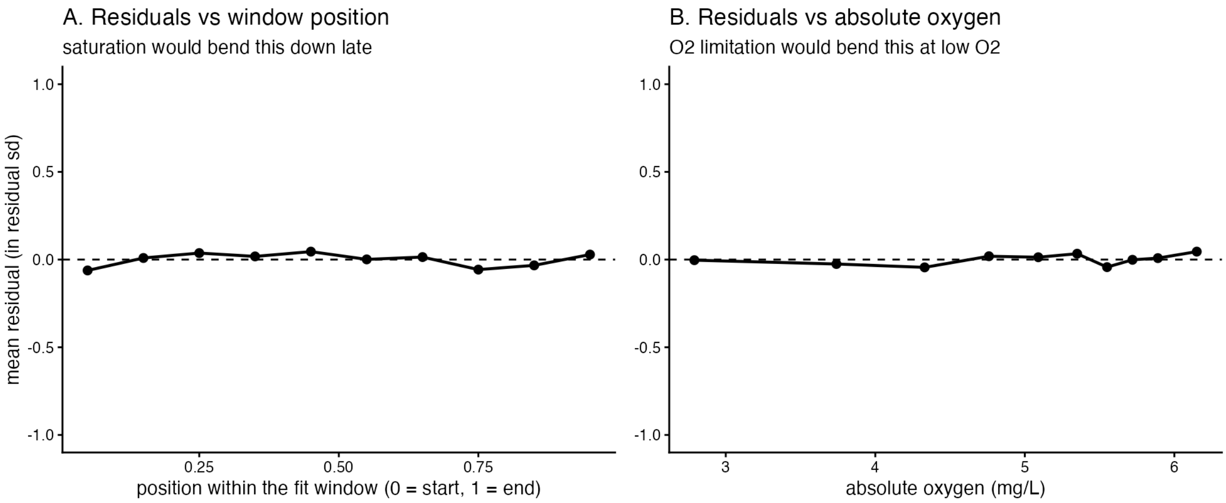

Figure 1: Neither signature is there. Saturation would bend panel A down late in the window; O limitation would bend panel B at low oxygen.

Table 2: Detectable deviation from the current model.

Test	Curves
nested F-test vs growth saturation, p < 0.05	3 / 75
nested F-test vs growth saturation, p < 0.01	1 / 75
nested F-test vs O2 limitation, p < 0.05	3 / 75
nested F-test vs O2 limitation, p < 0.01	0 / 75
|late-window mean residual| > 0.5 sd	0 / 75
|late-window mean residual| > 1.0 sd	0 / 75
window ends below 20% of starting O2	7 / 75

Method note, because the first attempt was wrong. A RESET-style regression of the residuals on the squared fitted value returned $p \geq 0.95$ for all 75 curves. That is not evidence of no effect — it is a degenerate test: for this three-parameter model $\hat{y}^2$ lies almost entirely in the span of the Jacobian, so the residuals are orthogonal to it by construction. It was discarded and replaced with a nested F-test against each saturating alternative, which is valid because M0 is the growth-saturation model at $N_{max} \to \infty$ and the O -limitation model at $K_m \to 0$. The counts above are from that test.

2.1 Why there is nothing to find

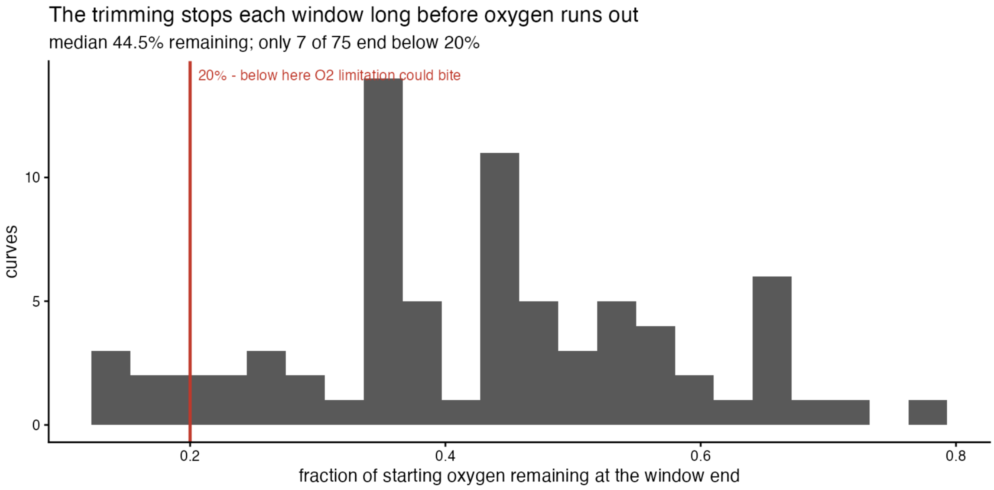

Figure 2

Table 3: The windows end well short of both failure modes.

Quantity	Min	Median	Max
O2 remaining at window end	13.8%	44.5%	77.8%
N-fold growth at window end	2.12	10.5	96.5
late-window mean residual (in sd)	-0.15	-0.014	0.14

`02_trimming.R` stops each window long before oxygen is exhausted. Bacterial respiration stays zero-order until O falls to a few percent of air saturation — far below where these windows end.

2.2 Mechanism discrimination is inconclusive, because there is no signal

Table 4: Neither mechanism's signature is favoured.

Predictor	Mechanism	corr with late residual	corr with curvature
min absolute O2 reached (mg/L)	O2 limitation	0.14	0.111
fraction of O2 remaining at window end	O2 limitation	0.143	0.111
elapsed growth r*t at window end	growth saturation	-0.0123	-0.222
N-fold increase at window end	growth saturation	0.000852	-0.226

Reported as indistinguishable, which is the honest answer when neither is present. Both are nevertheless implemented in §4, as the brief requires in that case.

3 The window-sensitivity hypothesis

`08_window_sensitivity.R` found K moving 15–34% under a ±6 min edge shift while r moves ~0.3%. The hypothesis was that this is saturation near the window end, and therefore fixable with a model term rather than tighter curation.

It is not.

Table 5: What predicts per-curve window sensitivity.

Predictor	Family	R^2 on max|dK|	R^2 on max|dR|
elapsed growth at end	saturation	0.764	0.132
fitted r	control	0.977	0.106
residual curvature	saturation	0.0597	0.00324
fraction O2 remaining	saturation	0.0886	0.001
residual sd	control	0.0374	0.000619
window length	control	0.174	0.000525
RSS gain growth-sat	saturation	0.00651	0.000348
late-window mean residual	saturation	0.00532	1.05e-05

All six saturation measures together reach $R^2 = 0.168$ (adjusted 0.095, $p = 0.045$). Fitted r alone reaches 0.977.

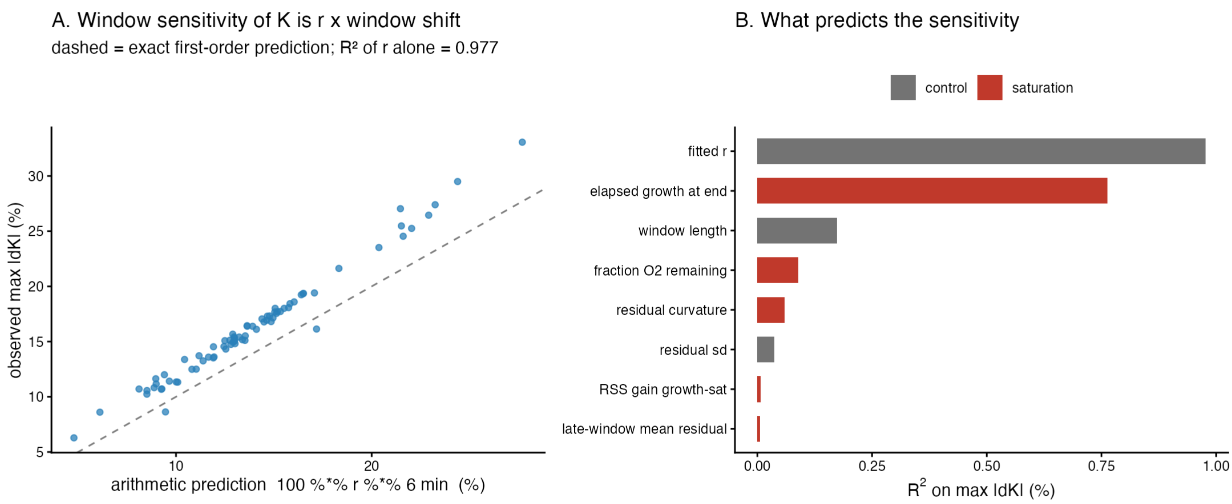

Figure 3: A: the observed sensitivity against the pure arithmetic prediction. B: saturation measures in red, controls in grey.

Table 6: The arithmetic explanation.

Quantity	Value
corr(max|dK_pct|, fitted r)	0.9885
R^2 of fitted r alone on max|dK_pct|	0.9772
median fitted r (per min)	0.02232
arithmetic prediction 100r6 (%)	13.39

Quantity	Value
observed median max|dK_pct| (%)	15.43
regression coefficient vs the arithmetic prediction (1.0 = exact)	1.169
R^2 of ALL saturation measures on max|dR_pct|	0.168

Shifting a window edge by Δ on a curve growing at rate r changes the oxygen consumed by $\approx e^{r\Delta}$, so K must move by $\approx r\Delta$ to compensate. **The window sensitivity is a mechanical property of fitting an exponential over a finite window** — it would be present in a perfectly specified model, and no model term removes it (§5).

The one predictor that looks like saturation — elapsed growth $r\,t_{end}$, $R^2 = 0.764$ — is r in disguise, since t_{end} varies far less than r across curves. It is shown next to r in the table so the confound is visible.

3.1 Correcting the brief's figures

"K and per-cell R by 15–34%" conflates the two. On `08`'s own basis:

Table 7: Per-curve window sensitivity from 08's committed table.

Quantity	Median	90th	Max
K	15.4%	24.1%	33.1%
per-cell R	2.2%	14.5%	21.1%
r	0.3%	0.8%	1.9%

Per-cell R is an order of magnitude less sensitive than K at the median, because $R = K\,O_{2,ref}/N_0$ and shifting the start moves N_0 in the same direction as K, partially cancelling.

4 The model variants

Six models per curve, as alternative fits. The default estimator is untouched.

M0	the pipeline's model (3 parameters)
M1	+ lag/equilibration t_{lag} (4)
M2o	+ O limitation, Michaelis–Menten K_m (4)
M2g	+ growth saturation, logistic N_{max} (4)
M3o / M3g	M1 + M2o / M1 + M2g (5)

M1 is a **nuisance** parameter, not biology: the paper's methods describe a ~45 min stabilisation during which the optode settles and the reported value drifts upward, and a lag term absorbs that instrumental transient.

Table 9: All 75 curves converge under every model.

Model	Converged	% at bounds	median AIC	median resid sd
M0	75/75	0.0%	-1109.56	0.001345
M1	75/75	41.3%	-1107.9	0.001348
M2g	75/75	0.0%	-1107.63	0.001345
M2o	75/75	40.0%	-1107.59	0.001342
M3g	75/75	40.0%	-1105.89	0.001355
M3o	75/75	68.0%	-1105.9	0.001346

Table 10: Head to head against M0.

Model	curves dAIC < -2	median dAIC	median dBIC	median RSS reduction	median shift in R
M1	3/75	2	4.59	-0.000%	-0.02%
M2g	3/75	1.99	4.49	0.011%	-0.34%
M2o	4/75	1.98	4.51	0.017%	-0.19%
M3g	4/75	3.92	9.17	0.075%	-1.43%
M3o	4/75	3.89	8.92	0.126%	-0.23%

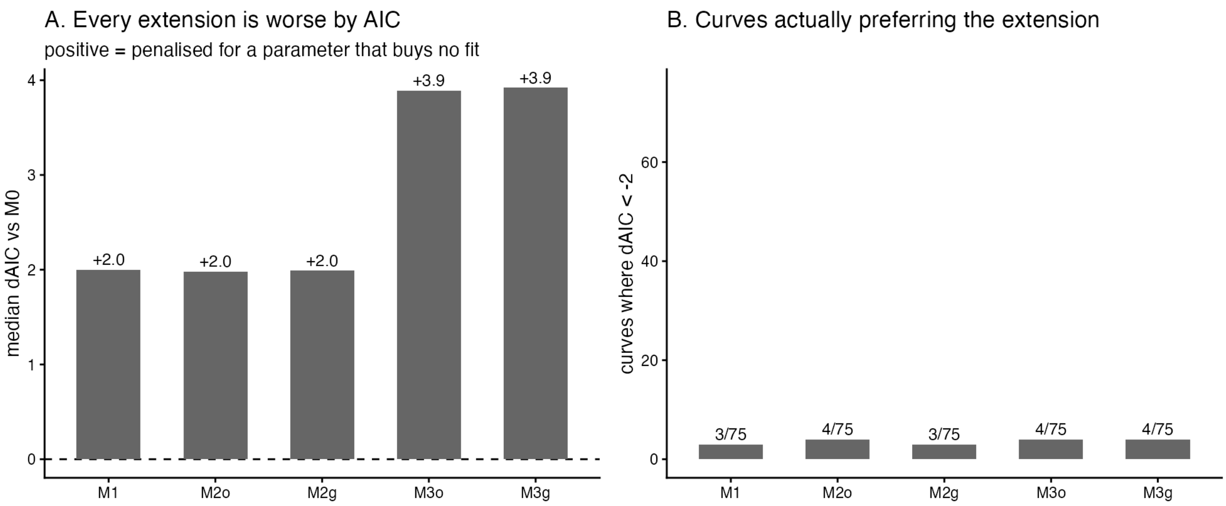

Figure 4

Median ΔAIC is **positive** and almost exactly the +2-per-parameter penalty — what happens when an extra parameter buys no fit at all. The extra parameters land where "no effect" lives:

Table 11: The extra parameters.

Model	Parameter	Median	Min	Max	% at bounds
M1	tlag (min)	0.1565	0	6.17	41.3%
M2g	Nmax_rel	2718	38.9	5.04e+05	0.0%

Model	Parameter	Median	Min	Max	% at bounds
M2o	Km (normalised O2)	0.002083	1e-06	0.119	40.0%
M3g	Nmax_rel; tlag	652.8	45.8	1.09e+05	40.0%
M3o	Km (normalised O2); tlag	0.002833	1e-06	0.112	68.0%

$N_{max} = 2718$ means biomass would increase that many fold before saturating, against an observed 10.5-fold over the window — the fitted "saturation" is placed far outside the data, which is M0 in the limit. Same for $K_m \to 0$ and $t_{lag} \to 0$.

4.1 No Bayesian escalation, and why

Parameters at bounds here are **not** an identification failure that a prior would fix. Every model converges, and the boundary *is* the answer: $t_{lag} \to 0$, $K_m \to 0$ and $N_{max} \to \infty$ all reduce the extended models to M0. A weak-prior Bayesian fit would return posterior mass piled at the same boundary and change no conclusion. Escalation is reserved for identification, and identification has not failed — D5's finding that uncertainty propagation is not the issue is not being reintroduced here as a justification.

5 The tests that matter

5.1 Power: can these models see the effects at all?

Re-using D5's misspecification generators, so simulated and observed diagnostics sit on the same footing.

Table 12: Power and specificity of the extensions.

Simulated truth	n	M1 wins	median t_lag	M2g wins	median N_max
lag	108	100.0%	15	100.0%	14.49
none	108	0.9%	0.04083	1.9%	319.8
saturating	108	17.1%	0.4449	47.2%	4.848

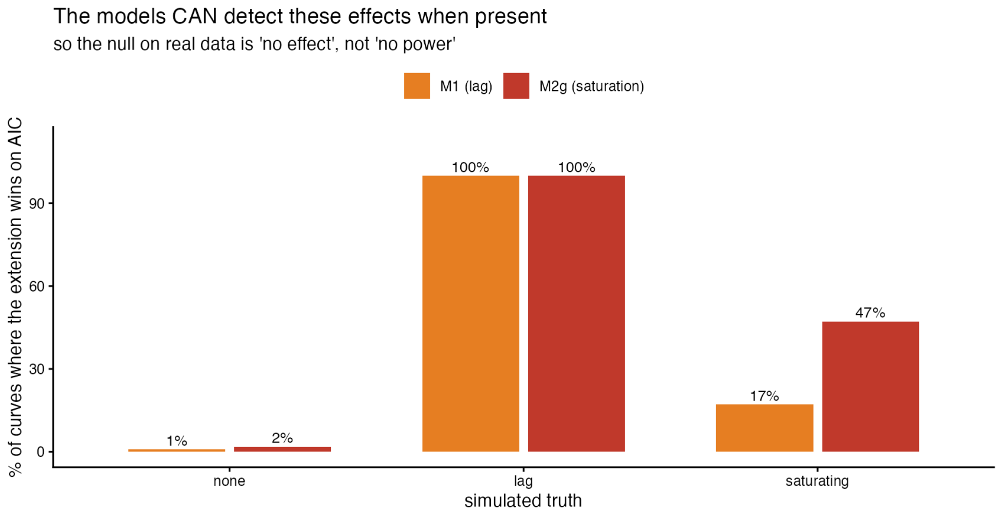

Figure 5

M1 recovers a planted 15-minute lag in 100% of cases at a median estimate of 15 minutes — exact — and is correctly rejected on clean data at 0.9%. **The null in §4 is "no effect", not "no power."** This is the load-bearing check for the whole report.

5.2 The key test: does window sensitivity fall?

Table 13: Window sweep, M0 vs M2g on identical windows.

Model	median max|dK|	p90 max|dK|	median max|dR|	median max|dr|
M0	14.45%	23.69%	14.67%	0.32%
M2g	14.53%	23.27%	15.34%	1.17%

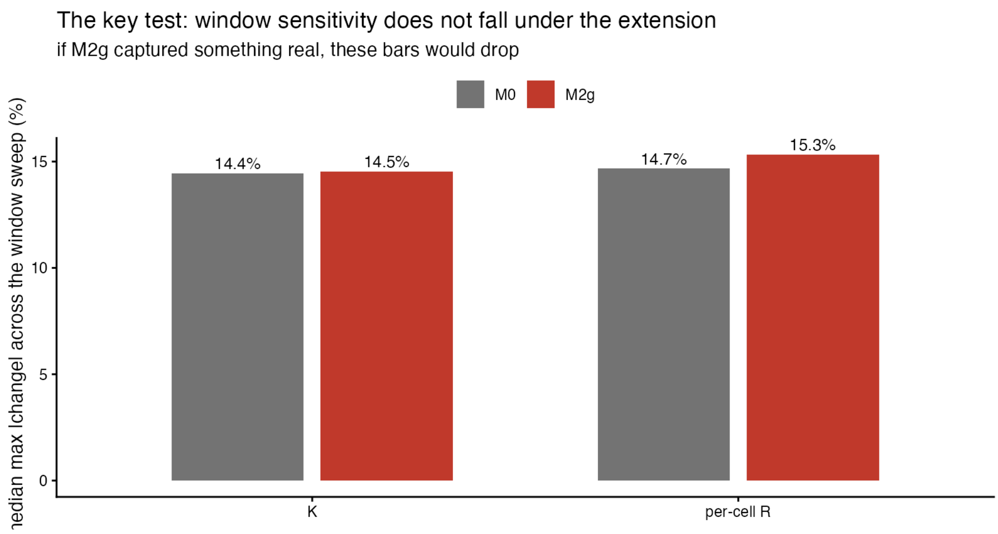

Figure 6

It does not. Sensitivity is flat for K and slightly *worse* for R and r. An extra parameter always improves likelihood, but if it were capturing something real the window sensitivity should drop.

Combined with §3's finding that the sensitivity tracks r at $R^2 = 0.977$, the picture is consistent: the sensitivity is arithmetic, and no model term removes it.

> **Caveat.** This sweep shifts edges within the already-trimmed `oxygen_fit_curves.csv`, so outward shifts are clipped where trimming removed points, while `08` re-trims from the raw series and can genuinely extend. The M0-vs-M2g comparison is valid — both models see identical windows — but the absolute percentages are not comparable with `08`'s committed table.

5.3 Downstream

Table 14: Between-taxon ordering and the growth-respiration relation.

Model	Spearman vs M0	median shift in R	pooled log-log slope	R^2
M0	1	0.00%	-0.000721	5.6e-08
M1	1	-0.06%	-0.00101	1.1e-07
M2g	1	-1.67%	0.0285	8.5e-05
M2o	1	-0.84%	0.00842	7.5e-06
M3g	1	-2.02%	0.0187	3.7e-05
M3o	1	-0.92%	0.00787	6.6e-06

Ordering is **identical** under every model. The pooled $\log R \sim \log r$ slope is ~0 throughout, consistent with D5's finding that the depletion anchor removes the R–growth coupling.

> This is the *pooled* log-log slope, **not** the published Fig 6 slope of 0.283, which `06_main_figures.R` obtains from a random-intercept mixed model. The comparison across models is like-for-like; the absolute value is not the published figure.

6 What it means

> **The current model is adequate for the follow-on work, and no extension is needed.** Both vulnerabilities D5 identified in simulation — an equilibration lag and a saturating growth phase — are absent from the real curves: a nested F-test rejects at chance level (3/75 and 3/75), the extra terms buy a median RSS reduction of ~0.01%, every extension is *worse* by AIC by almost exactly the per-parameter penalty, and the fitted parameters sit at the boundary where the extended model collapses back to the current one. This is not a power problem: the same models recover a planted 15-minute lag in 100% of simulated curves at a median estimate of 15.00 minutes. The reason the real curves are clean is the trimming — the median window still holds 44.5% of its starting oxygen when the fit ends, so neither oxygen limitation nor carrying capacity has been reached, and D5's −78.1% saturation arm was pessimistic about *this* data rather than wrong in general. The window sensitivity that motivated the hypothesis turns out to be arithmetic, not misspecification: it tracks the fitted growth rate at $R^2 = 0.977$ and matches the $r\Delta$ that shifting a window on an exponential requires, which is why no model term reduces it. **Nothing about the published estimates changes** — between-taxon ordering of per-cell respiration is identical under all six models (Spearman 1.000) and the median shift in per-cell R is under 2%. **Whether to change the default**

estimator is the repository owner's decision; on this evidence there is no case to.

6.1 Scope, and what would change this

- **This applies to the committed windows.** If trimming were relaxed to run closer to depletion, oxygen limitation would become reachable and the question should be reopened — 7 of 75 curves already end below 20% oxygen.
- **The lag question is about the trimming, not the model.** `02` starts each window at the oxygen tip, which already excludes the stabilisation transient. A lag term is redundant *given that trimming*, not in general.
- **D5's simulation arms remain valid as warnings.** They describe what would happen if these effects were present. D6 shows they are not, here.

Provenance

Table 15: Provenance. Environment fields come from env/versions.json.

Field	Value
Report	D6 — model structure
Git commit (environment record)	893e6d35c40134c1fb0853a499833e659d595ae3
Branch	gyd/packaging
Environment recorded (UTC)	2026-08-07 11:23:27 UTC
R	R version 4.5.2 (2025-10-31)
Platform	aarch64-apple-darwin20
OS	macOS Tahoe 26.2
renv lockfile	renv.lock — 158 packages, R 4.5.2
Artefact tree	runs/D6_analysis/
Seeds	simulation seed 20260807

Where every number came from

Table 16: All paths relative to runs/D6_analysis/.

Section	Artefact
§2 residual structure	A1_residual_structure_per_curve.csv, A2_..., A3_..., A4_deviation_counts.csv
§2.2 mechanism	A5_mechanism_discrimination.csv
§3 window hypothesis	B1_..., B2_correlations.csv, B3_variance_explained.csv, B4_..., B5_arithmetic_explanation.csv
§4 model variants	C1_model_fits_per_curve.csv, C2_model_summary.csv, C3_vs_M0.csv, C4_extra_parameters.csv
§5.1 power check	D1_power_check.csv
§5.2 window sweep	D2_window_sweep_raw.csv, D3_window_sweep_by_model.csv
§5.3 downstream	D4_ordering_and_fig6.csv, D5_headline.csv

Section	Artefact

Scripts are in `reports/D6_model_structure/scripts/`. No number was typed by hand; each is rendered from the CSV named above.